Now, according to the given information in question,

$$\frac{\frac{a-4}{2}}{4(b-4)}=\frac{5}{12} \text{ or } \frac{a-4}{2(4b-16)}=\frac{5}{12} \text{ or } \frac{a-4}{4b-16}=\frac{5}{6}$$

By cross multiplying we get

or, $6a - 24 = 20b - 80$

or, $6a - 20b = -56$

or $10b - 3a = 28$

After 8 years,

$$\frac{a+8}{2}+2=b+8$$

or, $\frac{a}{2}+4+2=b+8$

or, $b-\frac{a}{2}=-2$

or, $2b-a=-4$ ...(i)

or, $a=2b+4$ ....(ii)

Putting the value of a in equation (i), we get

$10b-3(2b+4)=28$

Or, $10b - 6b - 12 = 28$

Or, $4b = 40$

$\therefore$ $b = 10$

Hence, the present age of B is 10 years.

57. Let present ages of Simmi and Niti be a and b years, respectively.

Ten years hence, the ratio between Simmi's age and Niti's age = 7 : 9

According to the question, $\frac{a+10}{b+10}=\frac{7}{9}$

By cross multiplying we get

$\Rightarrow$ $9a+90=7b+70$

$\Rightarrow$ $7b-9a=20$ .....(i)

Also, $\frac{a-2}{b-2}=\frac{1}{3}$

By cross multiplying we get

$3a-6=b-2$

$\Rightarrow$ $3a-b=4$ ...(ii)

Multiplying equation (ii) by 3

$9a - 3b = 12$ ...(iii)

Adding equation (ii) and (iii) we get

$-9a+7b=20$

$4b = 32 \Rightarrow b = 8$year

From equation (ii) we get

$a=\frac{4+b}{3}=\frac{4+8}{3}=\frac{12}{3}=4$ years

Since, Abhay is 4 years older to Niti.

So, Abhay present age = 8 + 4 = 12 years

EXERCISE–B

(DATA SUFFICIENCY TYPE QUESTIONS)

Directions *(Questions 1 to 8):* ***Each of the questions given below consists of a statement and/or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is/are sufficient to answer the question. Read both the statements and***

Give answer (a) if the data in Statement I alone are sufficient to answer the question, while the data in Statement II alone are not sufficient to answer the question;

Give answer (b) if the data in Statement II alone are sufficient to answer the question, while the data in Statement I alone are not sufficient to answer the question;

Give answer (c) if the data either in Statement I or in Statement II alone are sufficient to answer the question;

Give answer (d) if the data even in both Statements I and II together are not sufficient to answer the question;

Give answer (e) if the data in both Statements I and II together are necessary to answer the question.

1. The sum of the ages of P, Q and R is 96 years. What is the age of Q ?

I. P is 6 years older than R.

II. The total of the ages of Q and R is 56 years.

2. What is Sonia's present age ?

I. Sonia's present age is five times Deepak's present age.

II. Five years ago her age was twenty-five times Deepak's age at that time.

3. How old is C now ?

I. Three years ago, the average of A and B was 18 years.

II. With C joining them now, the average becomes 22 years.

4. What is Reena's present age ?

I. Reena's present age is five times her son's present age.

II. Reena's age two years hence will be three times her daughter's age at that time.

5. What is the average age of A and B ? **(Bank P.O., 2007)**

I. The ratio between one-fifth of A's age and one-fourth of B's age is 1 : 2.

II. The product of their ages is 20 times B's age.

6. Average age of employees working in a department is 30 years. In the next year, ten workers will retire. What will be the average age in the next year ?

I. Retirement age is 60 years.

II. There are 50 employees in the department.